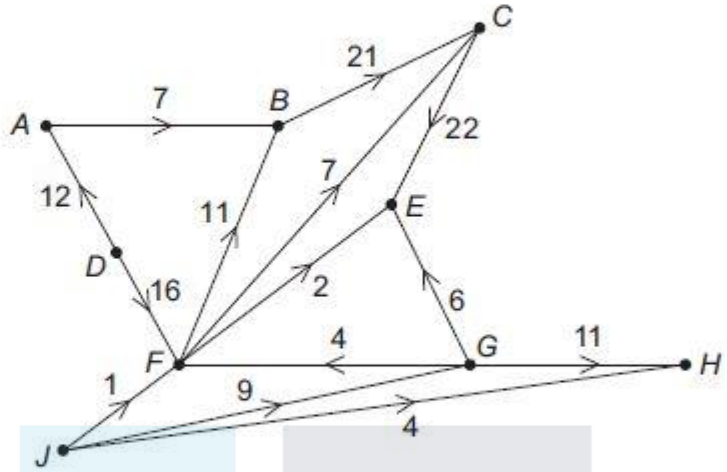
Q1.

The diagram shows a network of pipes.

Each pipe is labelled with its upper capacity in cm^3 s^{-1}



(a) (i) Find the value of the cut given by {*A*, *B*, *C*, *D*, *F*, *J*} {*E*, *G*, *H*}.

(1)

(ii) State what can be deduced about the maximum flow through the network.

(1)

(b) (i) List the nodes which are sources of the network.

(1)

(ii) Add a supersource *S* to the network.

(1)

(c) (i) List the nodes which are sinks of the network.

(1)

(ii) Add a supersink *T* to the network.

(1)

(Total 6 marks)

Q2.

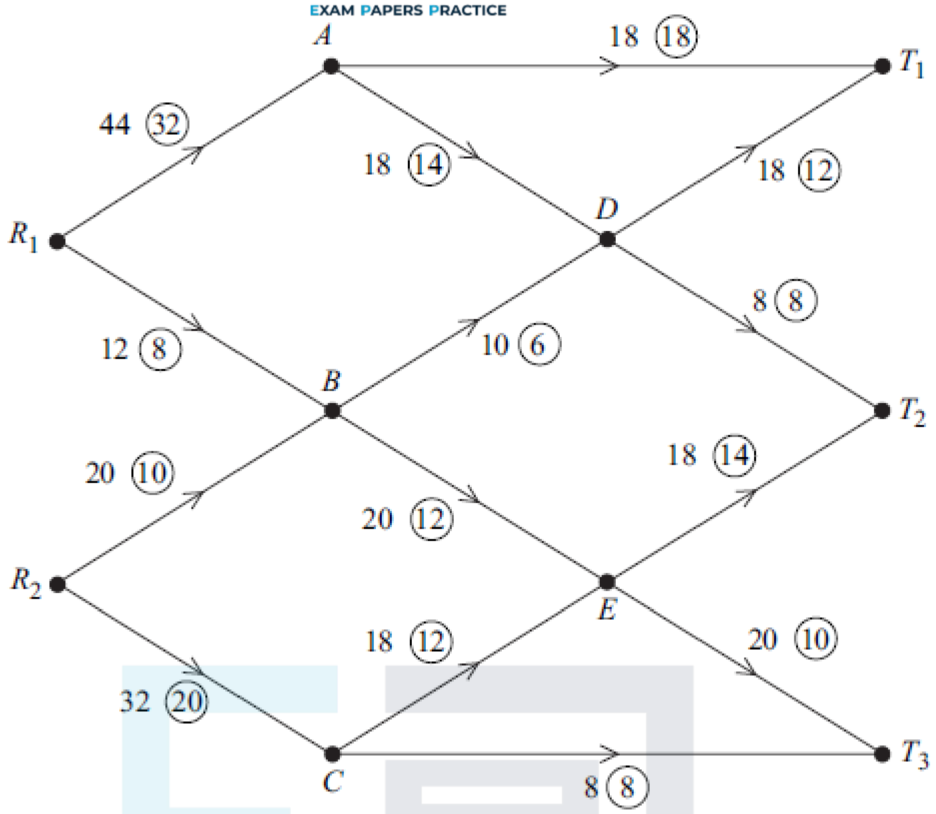
Figure 1 shows a network of pipes.

Water from two reservoirs, R_1 and R_2 , is used to supply three towns, T_1, T_2 and T_3.

In **Figure 1**, the capacity of each pipe is given by the number **not** circled on each edge.
The numbers in circles represent an initial flow.

(a) Add a supersource, supersink and appropriate weighted edges to **Figure 1**.

Figure 1

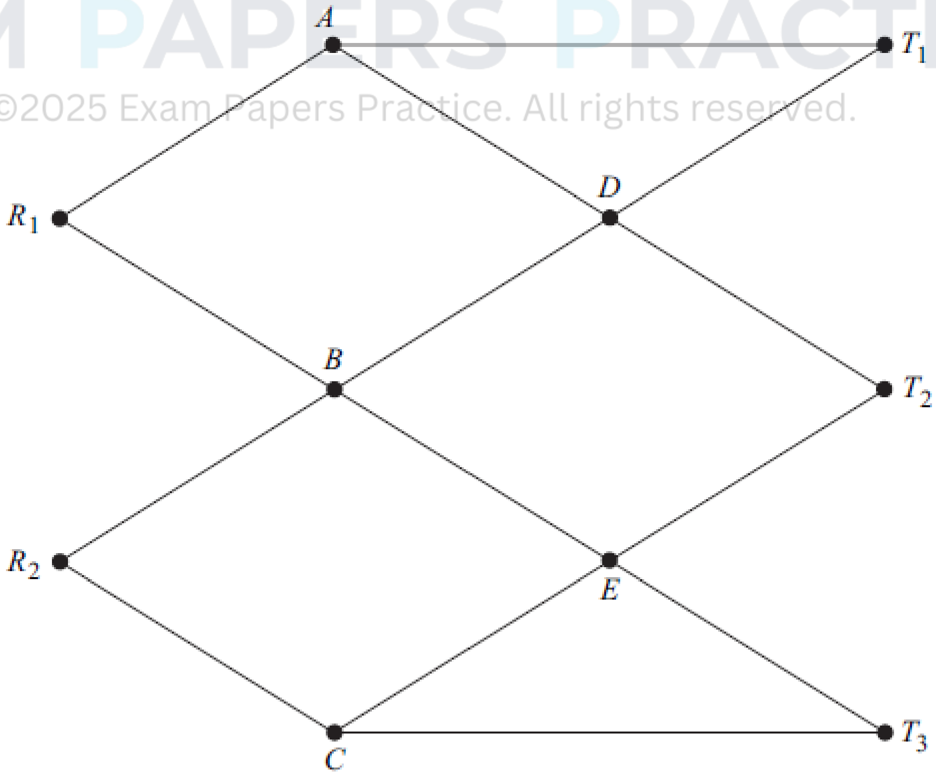


(2)

(b) (i) Use the initial flow and the labelling procedure on **Figure 2** to find the maximum flow through the network.

You should indicate any flow augmenting routes in the table and modify the potential increases and decreases of the flow on the network.

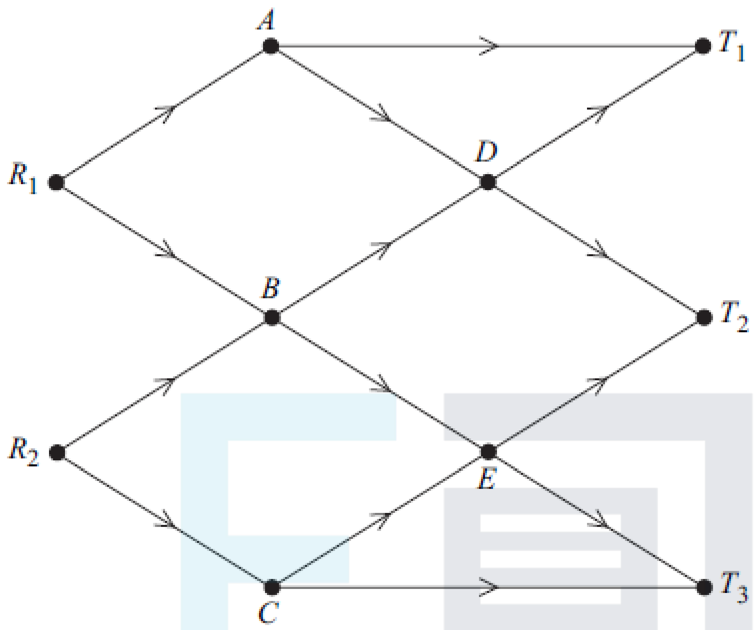
Figure 2



(5)

(ii) State the value of the maximum flow and, on **Figure 3**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 3



Route	Flow

(2)

(c) Confirm that you have a maximum flow by finding a cut of the same value. List the edges of your cut.

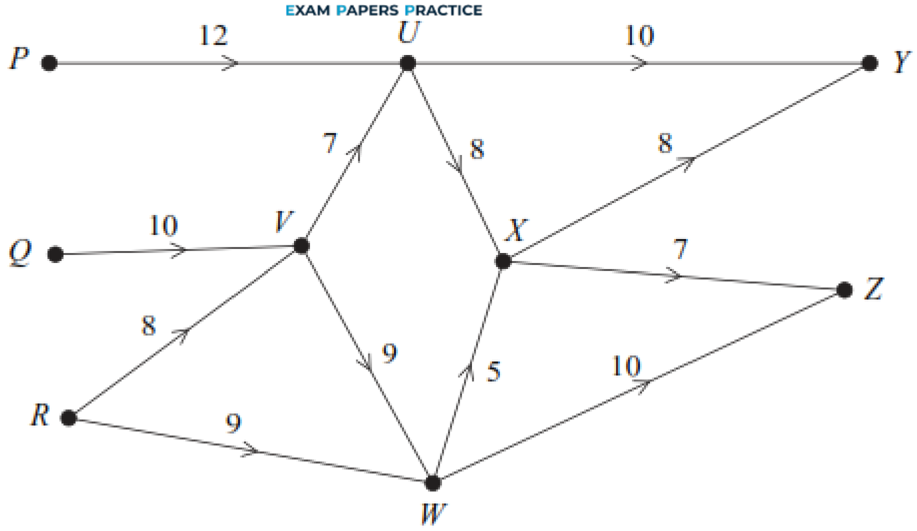
(2)

(Total 11 marks)

Q3.

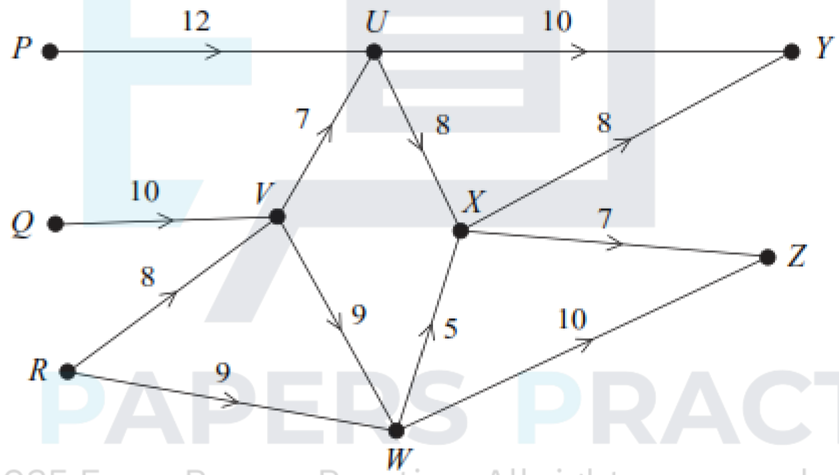
A retail company has warehouses at P, Q and R, and goods are to be transported to retail outlets at Y and Z. There are also retaining depots at U, V, W and X.

The possible routes with the capacities along each edge, in van loads per week, are shown in the following diagram.



(a) On **Figure 1**, add a super-source, S, and a super-sink, T, and appropriate edges so as to produce a directed network with a single source and a single sink. Indicate the capacity of each edge that you have added.

Figure 1



(2)

(b) On **Figure 2**, write down the maximum flows along the routes $SPUYT$ and $SRVWZT$.

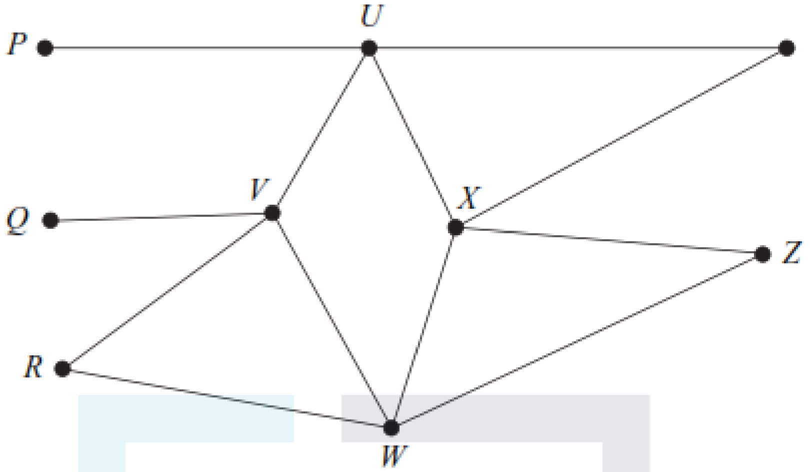
Figure 2

Route	Flow
$SPUYT$	
$SRVWZT$	

(2)

(c) (i) On **Figure 3**, add the vertices S and T and the edges connecting S and T to the network. Using the maximum flows along the routes $SPUYT$ and $SRVWZT$ found in part (b) as the initial flow, indicate the potential increases and decreases of the flow on each edge of these routes.

Figure 3



(2)

(ii) Use flow augmentation to find the maximum flow from S to T. You should indicate any flow-augmenting routes on **Figure 2** and modify the potential increases and decreases of the flow on **Figure 3**.

(4)

(d) Find a cut with value equal to the maximum flow.

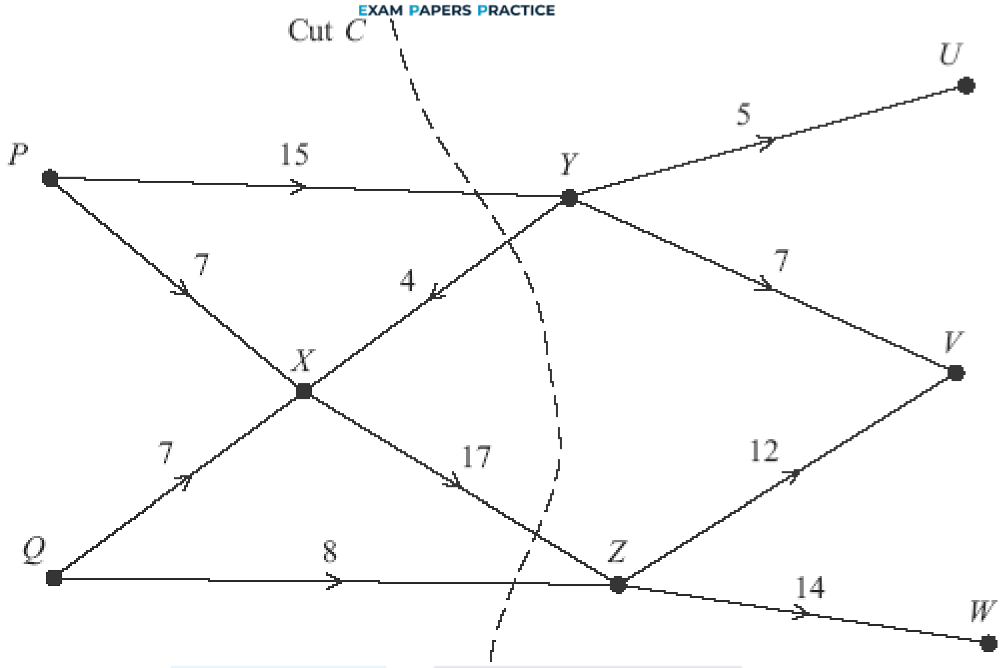
(1)

(Total 11 marks)

Q4.

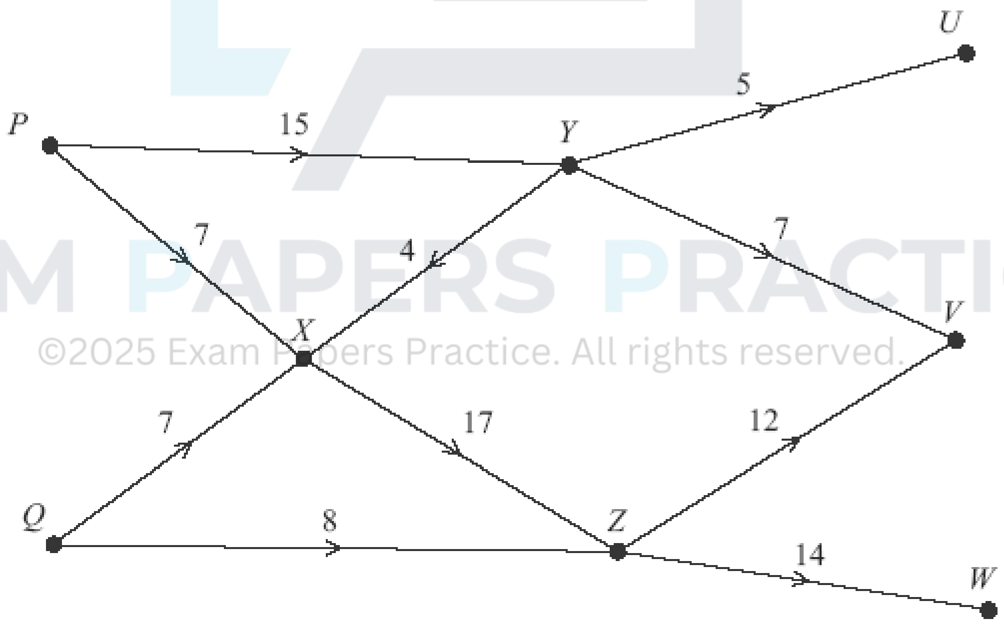
Water has to be transferred from two mountain lakes P and Q to three urban reservoirs U, V and W. There are pumping stations at X, Y and Z.

The possible routes with the capacities along each edge, in millions of litres per hour, are shown in the following diagram.



(a) On **Figure 1**, add a super-source, S, and a super-sink, T, and appropriate edges so as to produce a directed network with a single source and a single sink. Indicate the capacity of each of the edges you have added.

Figure 1



(2)

(b) (i) Find the value of the cut C.

(1)

(ii) State what can be deduced about the maximum flow from S to T.

(1)

(c) On **Figure 2**, write down the maximum flows along the routes $SQZWT$ and $SPYXZVT$.

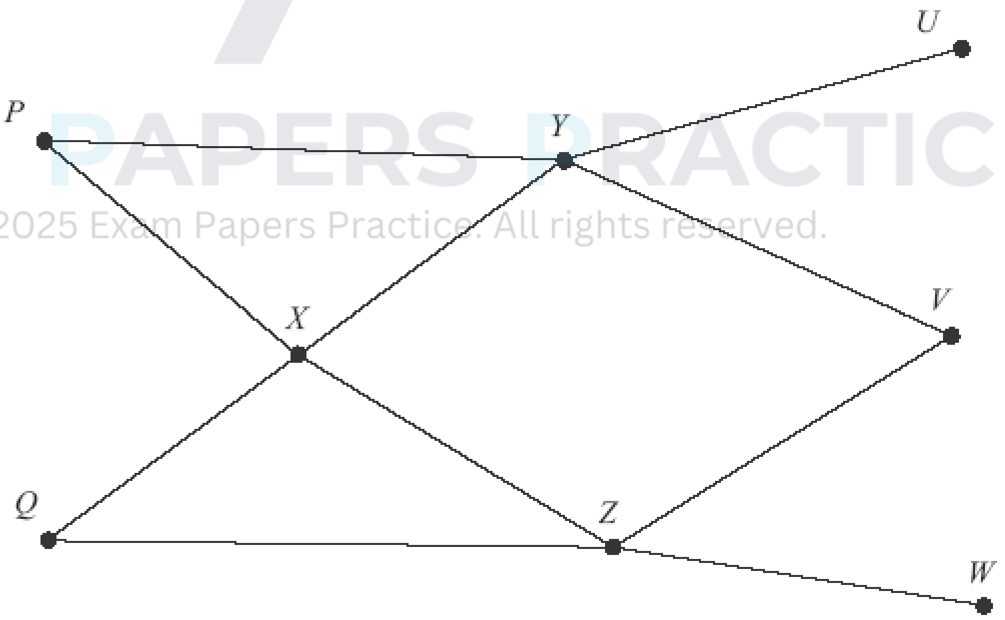
Figure 2

Route	**Flow**
$SQZWT$	
$SPYXZVT$	

(2)

(d) (i) On **Figure 3**, add the vertices S and T and the edges connecting S and T to the network. Using the maximum flows along the routes $SQZWT$ and $SPYXZVT$ found in part (c) as the initial flow, indicate the potential increases and decreases of flow on each edge.

Figure 3



(2)

(ii) Use flow augmentation to find the maximum flow from S to T. You should indicate any flow augmenting paths on **Figure 2** and modify the potential increases and decreases of the flow on **Figure 3**.

(4)

(e) State the value of the flow from Y to X in millions of litres per hour when the maximum flow is achieved.

(1)
(Total 13 marks)

Q5.

A greengrocer has two suppliers, A and B, and three storage depots, C, D and E. He needs to transport his stock to three retail outlets X, Y and Z. The capacities of the possible routes, in van loads per week, are shown in the following diagram.

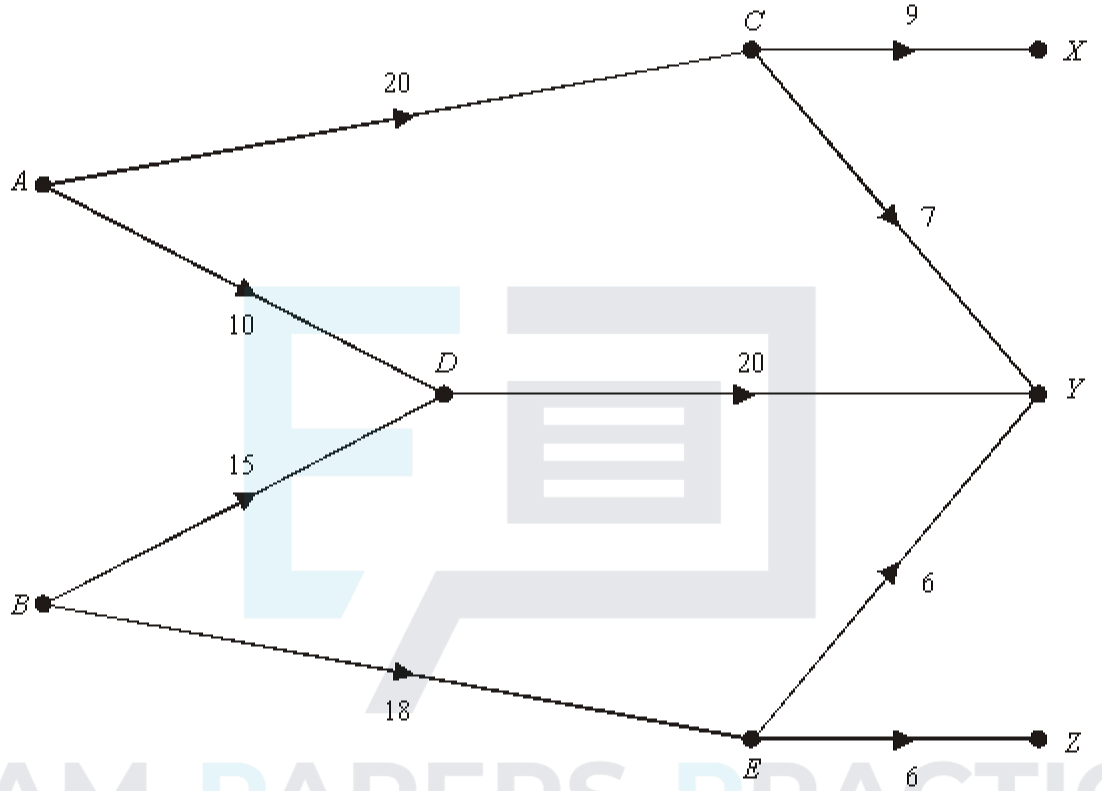


Figure 1

(a) Add a super-source (S) and a super-sink (W) on **Figure 1** to obtain a single source, single sink capacitated network. Show the capacities of each arc you have added.

(2)

(b) State the maximum flow along the routes $SADYW$ and $SBEZW$.

(2)

(c)

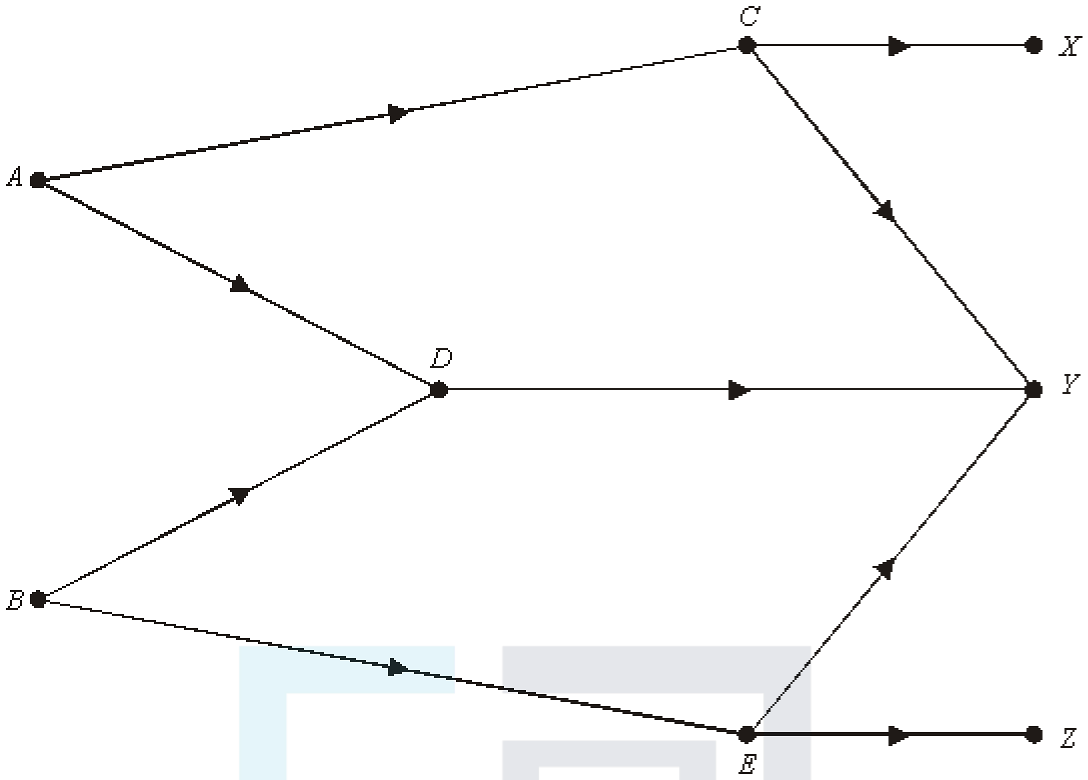


Figure 2

(i) Show your answers to part (b) on **Figure 2** and, taking this as the initial flow pattern, use flow augmentation to find the maximum flow from S to W.

(6)

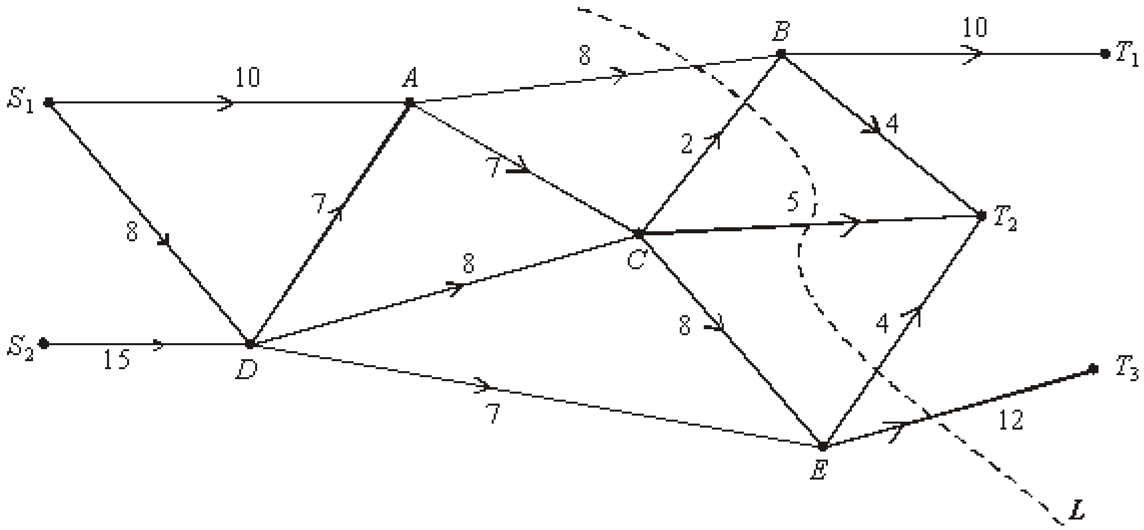
(ii) Prove that your flow is maximal.

(2)

(Total 12 marks)

Q6.

Two food factories S_1 and S_2 supply three major supermarkets T_1, T_2 and T_3. The various routes from factories to supermarkets are shown in the network below, where the number on each arc represents the daily maximum number of lorry-loads which can use that route.



(a) Add a supersource S and supersink T to the diagram above, together with appropriate arcs and capacities, in order to create an equivalent network with a single source and single sink.

(2)

(b) (i) Calculate the value of the cut shown by the broken line L above.

(1)

(ii) Find a cut of lower value than the one given in part (i).

(2)

(c) In your adapted network in the diagram above, start from a zero flow and use flow-augmenting paths to find a flow of value 30 from S to T.

(6)

(d) State how you can be sure that the flow found in part (c) is a maximum flow.

(2)

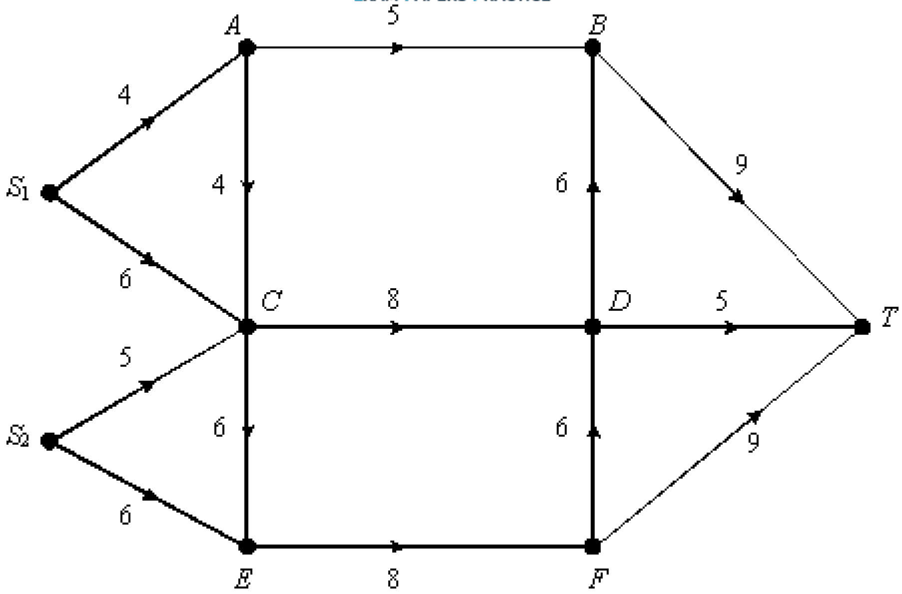
(e) Show that the three supermarkets cannot each be supplied with 10 lorry-loads in a day.

(3)

(Total 16 marks)

Q7.

In the network illustrated, S_1 and S_2 are sources, T is the sink, and the number on each arc indicates its capacity.



(a) On the diagram above add a super-source S and appropriate arcs and capacities in order to convert this network into one with a single source.

(2)

(b) By using flow augmentation, find a flow of 20 in this network. Mark the flow clearly on the diagram above.

(5)

(c) By finding an appropriate cut, explain how you know that a flow of 20 is the maximum possible.

(3)

(Total 10 marks)